TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Reactive Selective Laser Sintering of Thermoset Polymers for High-Performance Additive Manufacturing Applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	Composite Material Manufacturing Technologies	Plastic Deformation/ Powder	Powder Processing Technologies
Concept of Proposed Technology	The proposed technology aims to develop a reactive Selective Laser Sintering (SLS) process for the additive manufacturing of thermoset polymer components with significantly enhanced thermomechanical, chemical, and wear-resistance properties compared to conventional thermoplastic SLS parts. Conventional SLS processes rely on semi-crystalline thermoplastics (e.g., PA, PP, PPS, PEEK), which require high processing temperatures (>130°C) and often exhibit anisotropic mechanical properties, limited wear resistance, and thermal degradation during processing. These limitations restrict the use of SLS for high-load-bearing and long-term functional components. To overcome these constraints, the proposed approach introduces reactive polymeric materials (RPMs) based on non-isocyanate urethane-acrylate compounds, specifically engineered for laser-			

	induced curing during the SLS process. Unlike thermoplastics, these materials undergo in-situ polymerization and network formation, resulting in isotropic mechanical properties, improved chemical resistance, enhanced abrasion and creep resistance, and higher heat deflection temperatures. The technology further enables low-temperature SLS processing (50-80°C), significantly reducing energy consumption, thermal degradation risks, and overall process costs. Material properties can be systematically tuned through resin chemistry and filler selection. To ensure process stability and reproducibility, in-situ process monitoring systems are integrated to track temperature evolution, curing behavior, and microstructural development during layer-by-layer fabrication. The resulting platform enables the production of high-performance thermoset components for demanding applications, including biomedical implants and prosthetics.
Rationale for Joint R&D Project	The successful development of reactive SLS technology requires tight integration between materials chemistry, powder engineering, laser processing, process monitoring, and industrial validation, which cannot be achieved efficiently by a single organization. Joint R&D is essential because: • Reactive polymer design and scale-up require specialized expertise in non-isocyanate polymer chemistry and industrial synthesis. • Powder formulation and particle engineering must be optimized specifically for laser-based powder bed fusion processes. • SLS process parameter development and in-situ monitoring require advanced manufacturing and systems engineering capabilities. • Industrial demonstration and qualification are necessary to ensure commercial viability and application relevance. Collaboration with Korean SMEs enables: • Translation of laboratory-scale materials into manufacturable powder systems • Optimization of SLS equipment, process parameters, and quality control • Development of application-oriented demonstrators aligned with industrial and biomedical market needs • Acceleration of technology commercialization and global market entry Through international joint R&D, the project bridges advanced materials innovation and industrial additive manufacturing, positioning the technology for real-world deployment.
Key Words	1. Reactive Selective Laser Sintering 2. Thermoset Additive Manufacturing 3. Non-isocyanate urethane-acrylates 4. In-situ process monitoring 5. Biomedical implants

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5-6)
R&D Objectives	The overall objective is to establish a reactive SLS process platform enabling the additive manufacturing of high-performance thermoset components. Specific objectives include: • Development of reactive polymeric powder systems based on non-isocyanate urethane-acrylate chemistry • Optimization of powder morphology and particle size distribution for stable SLS processing • Reduction of SLS processing temperatures from >130°C to 50-80°C • Development of in-situ monitoring techniques to track curing, temperature, and microstructure evolution • Establishment of scalable synthesis, safety, and quality control protocols for RPM production • Demonstration of functional components for biomedical or other industrial applications			
R&D Target Specifications	• Processing temperature: 50-80°C • Powder particle size: 20-80 μm, near-spherical morphology • Mechanical properties: isotropic strength exceeding thermoplastic SLS benchmarks • Thermal performance: improved HDT and creep resistance • Process monitoring: real-time thermal and morphological observation • Scalability: RPM production demonstrated at ≥25 kg scale			
R&D Plan	The R&D activities are structured into the following phases: 1. Reactive polymer formulation development: Optimization of curing kinetics, crystallinity, and laser reactivity. 2. Powder engineering and characterization: Control of particle size, morphology, flowability, and packing density. 3. In-situ process monitoring development: Thermal imaging and subsurface observation of laser-induced curing. 4. Scale-up of reactive polymer production: Establishment of industrial synthesis, purification, and quality control protocols. 5. SLS process optimization: Definition of laser power, scan speed, layer thickness, and energy density. 6. Post-processing and material qualification: Mechanical, thermal, microstructural, and biological evaluation. 7. Demonstration and validation: Fabrication of biomedical implants			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Reactive polymer chemistry and formulation development • Powder characterization and curing kinetics analysis • Powder scale-up, granulation, and quality control • In-situ monitoring methodology development • Fundamental process understanding and validation • Application-specific performance validation • Data analysis and performance benchmarking • Data management and FAIR-compliant digital integration • Support for regulatory strategy and certification preparation
Expected R&R of Korean SME Partner	• Powder scale-up, granulation, and quality control • SLS equipment adaptation and process optimization • In-situ monitoring methodology development • Fundamental process understanding and validation • Prototype fabrication and industrial testing • Application-specific performance validation • Preparation for industrial scale-up and commercialization • Regulatory strategy and certification preparation for Korea/Asia

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated reactive SLS technology platform enabling the production of thermoset additively manufactured components with superior mechanical, thermal, and chemical performance. Expected impacts include: • Expansion of SLS beyond thermoplastics into high-performance thermoset materials • Reduction of energy consumption and processing costs through low-temperature SLS • New industrial opportunities in biomedical devices, prosthetics, and part consolidation • Strengthened the competitiveness of SMEs in advanced manufacturing and materials markets • Contribution to sustainable and high-value additive manufacturing technologies The technology aligns with global trends toward functional, durable, and application-ready additive manufacturing solutions.

6. References

(1) 1 K. Wudy and D. Drummer, *JOM*, 2019, **71**, 920–927.

(2) W. Zhu, C. Yan, Y. Shi, S. Wen, J. Liu, Q. Wei and Y. Shi, *Sci. Rep.*, 2016, **6**, 33780.

(3) N. Singh, H. Bakhshi and W. Meyer, *RSC Adv.*, 2020, **10**, 44103–44110.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization